Final Project Deep Learning

Using the Nutrition5k Dataset | 3 Model Architecture Comparison

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Project Definition

This project tackles the problem of automated nutrition estimation from food images.

Given a single overhead photograph of a prepared dish, the model predicts five nutritional values — calories, mass, fat, carbohydrates, and protein.

https://github.com/google-research-datasets/Nutrition5k

Dataset: Nutrition5k

https://github.com/google-research-datasets/Nutrition5k

5,006 unique dishes — each assembled in a lab with known ground-truth weights

Overhead RGB images + 4× side-angle video streams

~3,500 dishes have overhead images (subset used here)

Targets (continuous regression): Calories · Mass · Fat · Carbs · Protein

Split used in this project:

Train: 4,058 dishes Test: 708 dishes

Label normalization applied per-target (zero mean, unit variance) during training

Augmentation: random horizontal flip + random crop

Approach: Shared Backbone + Task Heads

Backbone: InceptionV3 pretrained on ImageNet

— Paper used InceptionV2 + JFT-300M (not publicly available)

Shared representation: 2048-dim feature vector from backbone

Multi-task regression: 5 independent (or conditioned) output heads

Loss: L1 (Mean Absolute Error) — robust to outliers

Optimizer: RMSProp (lr = 0.001, momentum = 0.9)

Schedule: CosineAnnealingLR (T_max = 100 epochs)

Backbone frozen for first 5 epochs, then fine-tuned end-to-end

Model 1: Baseline CNN

Architecture:

InceptionV3 (2048) → Dropout(0.6) → FC(4096) → ReLU

→ 5 × FC(4096) → ReLU → FC(1) [one head per target]

5 completely independent prediction heads

Direct translation of the paper's described architecture

Training: batch=32, 100 epochs, early stopping patience=15

Model 2: Mass-Focus

Hypothesis: Mass is the most important nutrient; All other nutrients scale with mass.

Architecture:

InceptionV3 (2048) → mass_head → predicted_mass

[2048 || predicted_mass] (2049) → 4 × FC(4097) → ReLU → FC(1)

[cal, fat, carbs, protein conditioned on mass]

Task weights: TASK_WEIGHTS = [1, 3, 1, 1, 1] (mass weighted 3×)

Extra loss term encourages accurate mass prediction before downstream heads

Model 3: Mixture of Experts (MoE)

Architecture:

InceptionV3 (2048) → Gate Network (2048 → 4 logits, softmax)

4 Expert Networks: each FC(2048) → ReLU → FC(2048) → ReLU → FC(2048)

Weighted sum of expert outputs → 5 Task Heads: FC(2048→1024) → ReLU → FC(1)

Gating encourages specialization: different experts handle different dish types

Soft routing – all experts run, weighted by gate score

Training Details (All Models)

Common hyperparameters:

- Batch size: 32
- Max epochs: 100
- Learning rate: 0.001
- Dropout: 0.6
- Backbone freeze: first 5 epochs
- Early stopping: patience = 15

- Optimizer: RMSProp (momentum=0.9)
- Scheduler: CosineAnnealingLR
- Loss: L1 (MAE)

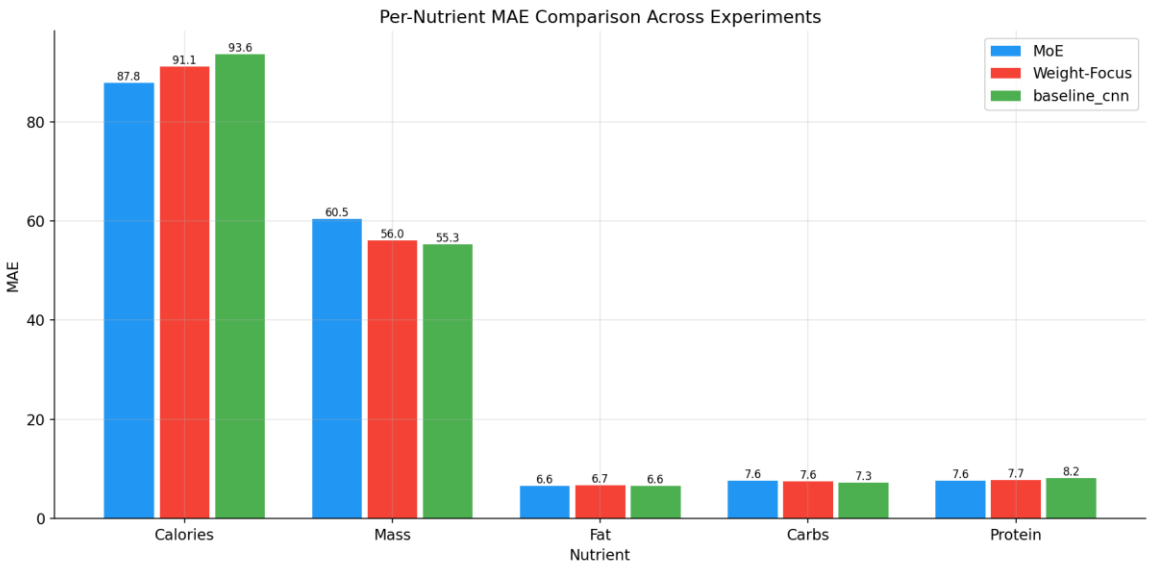
Label normalization:

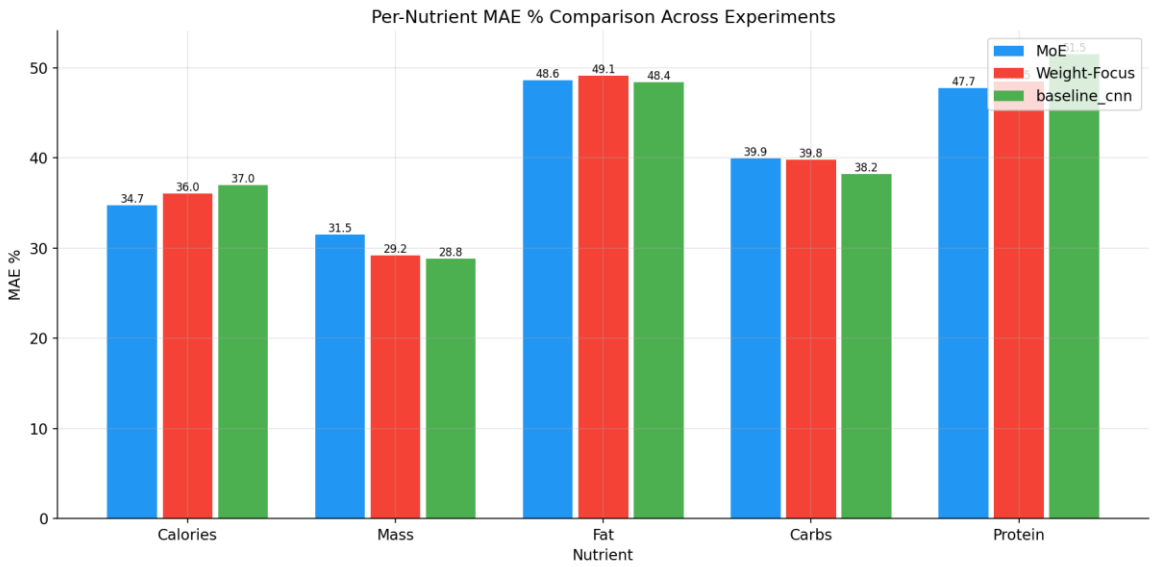
- Mean & std computed on training set
- Applied per target independently
- Predictions de-normalized for evaluation

Data augmentation:

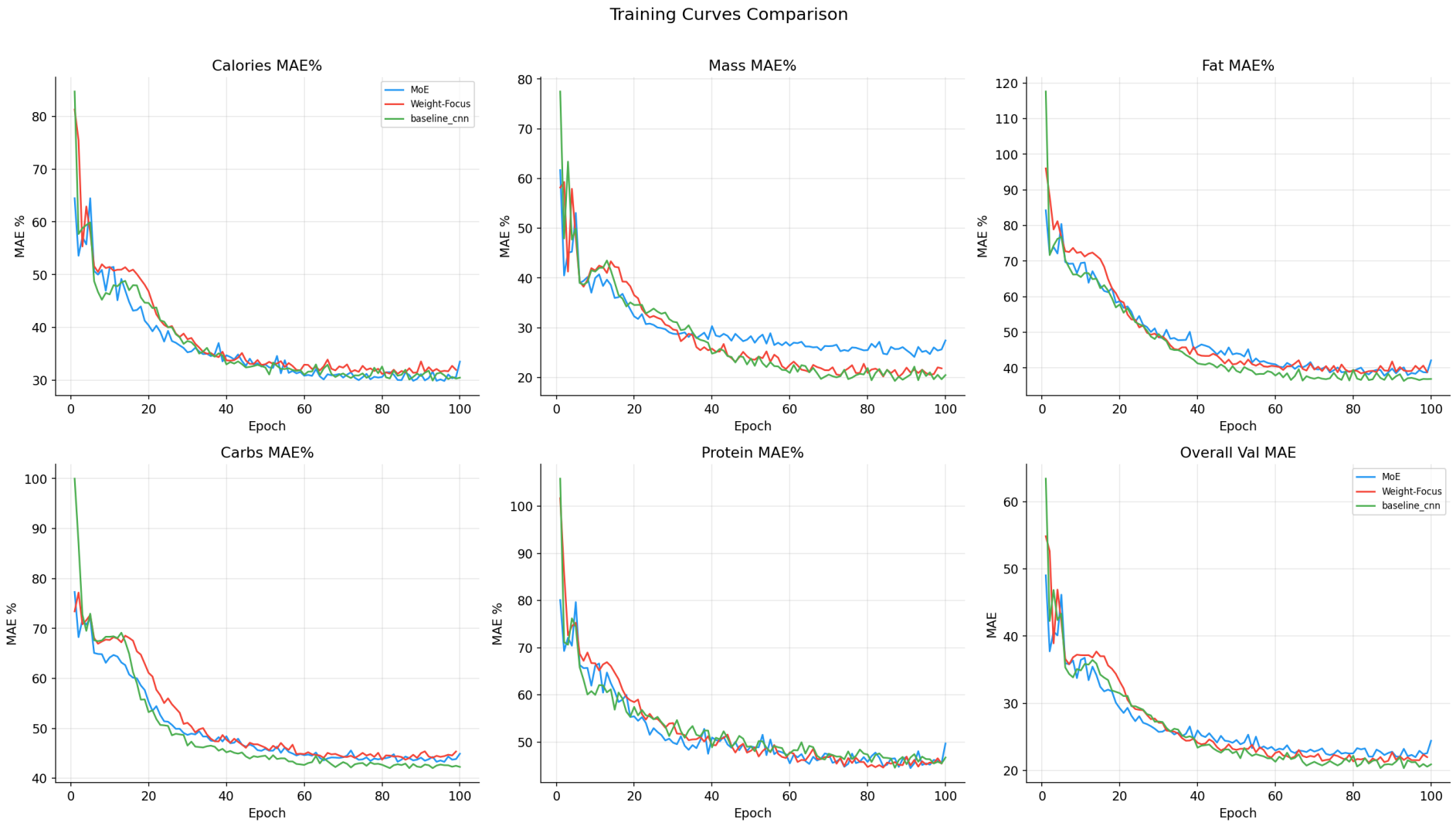
- RandomHorizontalFlip
- RandomResizedCrop(299)
- Normalize (ImageNet stats)

Results: MAE Comparison

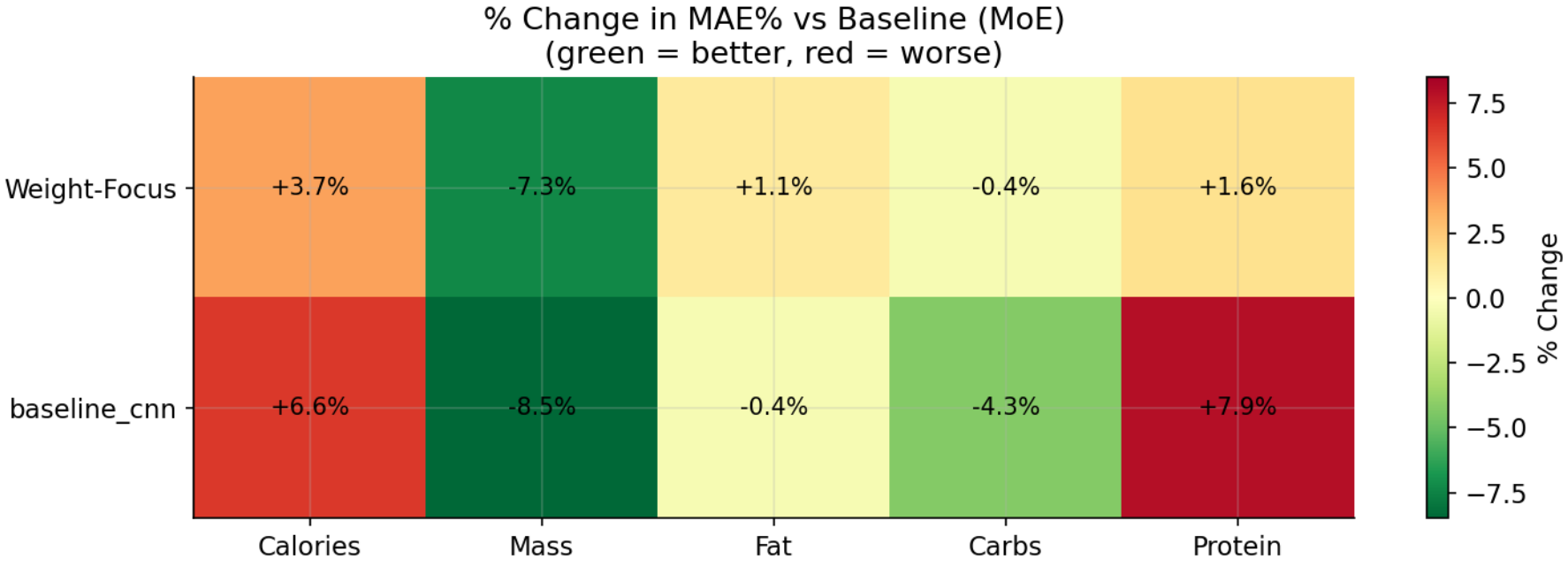




Results: Training Curves



Results: Improvement Heatmap vs MoE



Conclusions

- Baseline CNN achieved the best MAE% (Simple multi-head regression is hard to beat)
- Weight Focus added complexity without the consistent gain
    - Mass Conditioning didn’t reliably help other targets as much as expected
- MoE halves parameter count (65.9M vs 130.9M) with slight MAE% error.
    - Promising efficiency trade-off for deployment.

What else to explore:

- More aggressive data augmentation.
- Larger / newer backbone (EfficientNet-V2, etc.)

Questions?